

# FORENSICSHIELD COMPLIANCE REPORT

Report ID: RPT-20260913-3555A51B  
Case #: CAS-SIH-2026-001

## 1. Executive & Metadata Summary

|               |                                               |                     |                         |
|---------------|-----------------------------------------------|---------------------|-------------------------|
| Case Title:   | Smart India Hackathon 2026 Demonstration Case | Generated At:       | 2026-09-13 08:55:31 UTC |
| Case Number:  | CAS-SIH-2026-001                              | Generated By:       | admin                   |
| Tool Version: | ForensicShield v1.0.0                         | Audit Chain Status: | INTACT                  |

## 2. Forensic Status Classification Breakdown

| Status Category | Verified | Inconclusive | Failed | Unsupported | Manual Review |
|-----------------|----------|--------------|--------|-------------|---------------|
| Item Count      | 4        | 1            | 0      | 0           | 1             |

## 3. Read-Only Evidence Image Inventory

| Evidence ID     | Item #   | Original Name        | Size (Bytes) | SHA-256 Digest      | Status   |
|-----------------|----------|----------------------|--------------|---------------------|----------|
| EVD-SIH-DISK-01 | ITEM-001 | synthetic_test_disk_ | 10,485,760   | effcb0dbb243905b... | VERIFIED |

## 4. Recovered & Carved Forensic Artifacts

| Artifact ID     | Type / Format | Source Offset | SHA-256 Digest      | Confidence / Status |
|-----------------|---------------|---------------|---------------------|---------------------|
| CARV-JPEG-001   | Carved JPEG   | 10,240        | hash_carv-jpeg-0... | Confidence: HIGH    |
| CARV-PNG-001    | Carved PNG    | 150,000       | hash_carv-png-00... | Confidence: HIGH    |
| CARV-PDF-001    | Carved PDF    | 300,000       | hash_carv-pdf-00... | Confidence: HIGH    |
| CARV-ZIP-001    | Carved ZIP    | 500,000       | hash_carv-zip-00... | Confidence: HIGH    |
| CARV-JPEG-TRUNC | Carved JPEG   | 800,000       | hash_carv-jpeg-t... | Confidence: LOW     |

## 5. Technical Limitations & Forensic Scope

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**MANDATORY FORENSIC BOUNDARIES & TECHNICAL DISCLAIMERS:**

- 1. Storage Media Differences:** Magnetic HDDs support direct block overwriting. SSDs and NVMe drives utilize Flash Translation Layers (FTL), wear-leveling algorithms, and over-provisioned blocks. Logical sector overwriting does not guarantee sanitization of unmapped physical NAND blocks.
- 2. Filesystem Boundaries:** Filesystem recovery is restricted to supported structures (FAT32, NTFS, Ext4). Damaged or unknown volumes require manual review.
- 3. Signature Carving & Fragmentation:** Pure signature carving assumes contiguous block allocation. Carving non-contiguous fragmented files without allocation metadata produces corrupt byte streams.
- 4. Validation Scoring Heuristics:** Validation scores (0-100) represent an empirical heuristic structural index and are NOT statistical probabilities of truth.
- 5. Compliance Phrasing Notice:** Absolute claims ('100% unrecoverable') are excluded under ISO/IEC 27037 standards.